

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Fuel System Clean Care Kit, Part Number 4919073
- QD™ contact cleaner, Part Number 3824510, or equivalent

Additional Service Items

- No additional service items required.

Preparatory Steps

Recommended Cummins® Service Tools

- Fuel System Clean Care Kit, Part Number 4919073
- QD™ contact cleaner, Part Number 3824510, or equivalent

Additional Service Items

- No additional service items required.

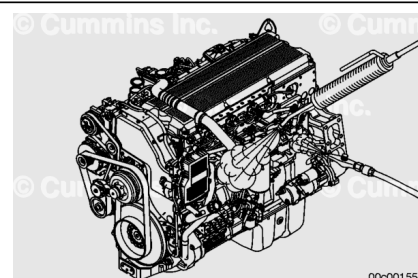
⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠



Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.

Before servicing any fuel system components, (such as fuel lines, fuel pump, injectors, etc.), which would expose the fuel system or internal engine components to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine. Refer to Procedure 000-009 (/qs3/pubsys2/xml/en/procedures/99/99-000-009.html) in Section 0.

To prevent damage from debris and contamination, cover, cap, or plug any openings as soon as possible when servicing the fuel system. Caps and plugs can be found in Fuel System Clean Care Kit, Part Number 4919073.

⚠ WARNING ⚠

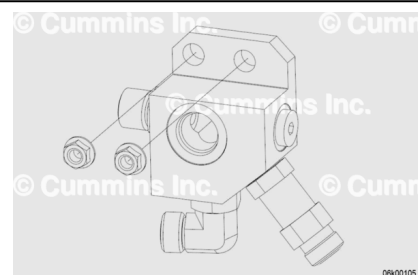
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the vehicle fuel return line from the fuel drain manifold. See equipment manufacturer service information.
- Disconnect the fuel drain lines from the fuel drain manifold. Refer to Procedure 006-013 (/qs3/pubsys2/xml/en/procedures/20/20-006-013-tr.html) in Section 6.

Remove

Remove the two nuts securing the fuel drain manifold to the engine.

Remove the fuel drain manifold.



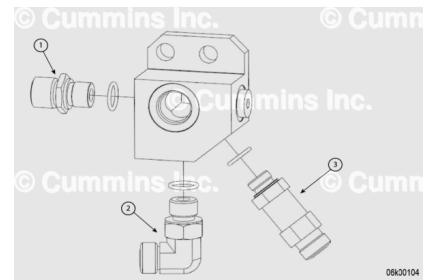
Disassemble

The fuel drain manifold has three fittings:

1. Injector return check valve
2. Pressure limiting valve drain line
3. Low-pressure air bleed line check valve

Remove the fittings from the fuel drain manifold.

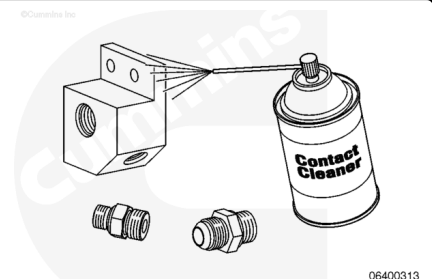
Discard the o-rings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



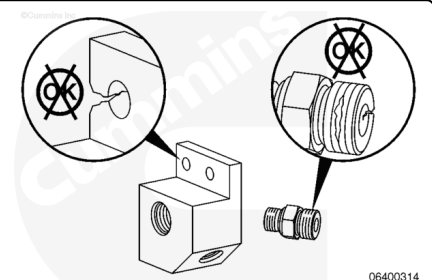
⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Clean all parts. Use QD™ contact cleaner, Part Number 3824510, or equivalent.

Inspect the fuel manifold for cracks, stripped threads, and other damage. Replace, if damaged.

Inspect the union fittings and check valves for cracks, stripped threads, and other damage. Replace, if damaged.



Assemble

Lubricate all new o-rings with vegetable oil prior to installation.

Install the injector return check valve (1) with new o-rings into the fuel drain manifold.

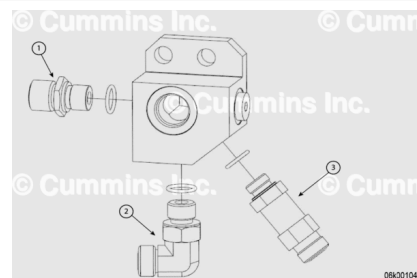
Torque Value: 54 n•m [40 ft-lb]

For options with a male union elbow, install the pressure limiting valve union fitting (2) with new o-rings into the fuel drain manifold.

Torque Value: 47 n•m [35 ft-lb]

Install the low-pressure air bleed check valve (3) with new o-rings into the fuel drain manifold.

Torque Value: 37 n•m [27 ft-lb]

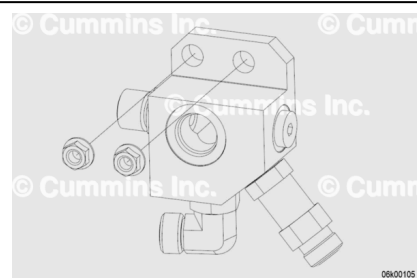


Install

Install the fuel drain manifold to the engine.

Tighten the two nuts.

Torque Value: 27 n•m [239 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the fuel drain lines. Refer to Procedure 006-013 (</qs3/pubsys2/xml/en/procedures/20/20-006-013-tr.html>) in Section 6.
- Connect the fuel return line to the fuel drain manifold. See equipment manufacturer service information.

- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

Last Modified: 09-Aug-2022
